II. Elementary Probability Theory (GSE 518)

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In order to understand the concepts and methods that we will discuss in this class, it is important to have an underlying understanding of some key concepts and ideas from probability theory.

1. Elementary Probability Theory

There are *three* basic elements of probability theory: Sample Space, Events, and Probability Function.

1) Sample Space

Definition:

The sample space S is the set of all possible outcomes of an experiment.

Example: Flip 2 coins

$$S_1 = \{HH, HT, TH, TT\}$$

Example: Toss a die

$$S_2 = \{1, 2, 3, 4, 5, 6\}$$

2) Event

Definition:

An event E is defined as any subset of S.

Example:

Let S be the sample space for tossing a die

$$E_1 = \text{get an odd number} = \{1, 3, 5\} \subseteq S$$

$$E_2 = \emptyset \subseteq S$$

$$E_3 = S$$

3) Probability Function

Preliminary Ideas:

A probability function is a function that assigns probabilities to events E, typically defined as $P(E)$. A probability function's role is to assign to each subset of S a unique number between 0 and 1 (a probability) $P(E)$. Another way to say this is the probability function takes in sets and outputs real numbers between 0 and 1.

Note that $P(E)$ is a "set function", meaning its domain is made up of sets. This domain is not always easy to define as the domain may be either countable or uncountable. We will let $\mathcal{B}$ be the Domain of the probability function given by:

Case 1: If S is finite or countable then the Domain can be defined as the set of all possible subsets of S. Thus $\beta = \{\text{all subsets of S}\}$

Example: Flip one coin $\implies$ $S = \{H, T\}$. (this is a discrete and thus countable set)

Then then the Domain of the probability function would be $\{H, T, S, \emptyset\}$

Case 2: If S is uncountable, we will define the Domain of the probability function as all intervals of $\mathbb{R}$. Thus $\mathcal{B} = \{\text{all intervals of } \mathbb{R}\}$.

$\mathcal{B}$ is called a "Borel field".

Formal Definition of a Probability Function and Axioms of Probability:

We define $P : \mathcal{B} \to \mathbb{R}$ as a *probability function* on $(S, \mathcal{B})$ where S is the sample space and $\mathcal{B}$ is the domain, if it satisfies the following:

(a) $P(A) \geq 0 \quad \forall A \in \mathcal{B}$ (A is an event in the domain)

(b) $P(S) = 1$

(c) If $A_i \in \mathcal{B}$ for $i \in \mathbb{N}$ are pairwise disjoint, then:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

a to c are known as the axioms of probability or Kolmogorov's axioms. The triple $(S, \mathcal{B}, P)$ is called the probability space.

2. Implications of Kolmogorov's Axioms

1. $P(A^c) = 1 - P(A)$ for any $A \in \mathcal{B}$

Proof:

$S = A \cup A^c$

Note that $1 = P(S)$ by axiom b.

Then:

$1 = P(S) = P(A \cup A^c)$

Note that $A \cap A^c = \emptyset$.

Then by axiom c we have that

$P(A \cup A^c) = P(A) + P(A^c)$

$\Rightarrow 1 = P(A) + P(A^c)$

$\Rightarrow P(A^c) = 1 - P(A)$

2. $P(A) \leq 1$

Proof:

$0 \leq P(A) = 1 - P(A^c)$

$\longrightarrow$ by axiom a, and the result above in 1).

$\longrightarrow$ Also note that since $P(A^c) \geq 0$, then $P(A) \leq 1$

3. $P(\emptyset) = 0$

Proof:

$1 = P(S) = P(S \cup \emptyset) = P(S) + P(\emptyset)$

$S \cap \emptyset = \emptyset$

but $P(S) = 1$, so $1 = 1 + P(\emptyset) \Rightarrow P(\emptyset) = 0$

4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Proof:

From the "concluding results" section from our discussion of sets (proofs #3 and #1) that

$A \cup B = A \cup (B \cap A^c)$

$\Rightarrow P(A \cup B) = P(A) + P(B \cap A^c)$

We also know

$B = (A \cap B) \cup (A^c \cap B) \Rightarrow P(B) = P(A \cap B) + P(A^c \cap B)$

Finally, combine the equations for $P(A \cup B)$ and $P(B)$ obtained above (e.g., solve for $P(A^c \cap B)$ in one equation and plug into the other) to get:

$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$

5. If $A \subseteq B$ then $P(A) \leq P(B)$

Proof:

Let $A \subseteq B$.

Note that we already showed:

If $B \subseteq A$ then $A = A \cap B \cup B$

Thus we have that

$B = B \cap A \cup A$

$\Leftrightarrow B = (B \cap A^c) \cup A$

$\Leftrightarrow P(B) = P(B \cap A^c) + P(A)$

$\longrightarrow$ since $(B \cap A^c) \cap A = \emptyset$

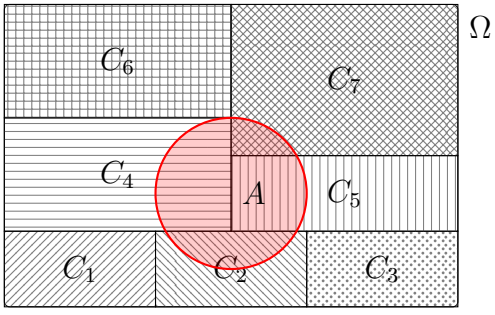
We know $P(B \cap A^c) \geq 0$ from our axioms.

$\Rightarrow P(B) - P(A) = P(B \cap A^c) \geq 0 \Rightarrow P(B) \geq P(A)$ as desired.

6. For any partition $C_1, C_2, C_3, \ldots$ of S Then for any $A \in \mathcal{B}$:

$$P(A) = \sum_{i=1}^{\infty} P(A \cap C_i)$$

Intuition:



$P(A) = \sum_{i=1}^{7} P(A \cap C_i)$

Example:

$S = \{\text{all individuals in population}\}$

$C_1 = \{\text{males}\}, C_2 = \{\text{females}\}$

$A = \{\text{white}\}$

$P(A) = P(\text{white males}) + P(\text{white females})$

7. *Boole's Inequality*:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} P(A_i)$$

Note: that $\sum_{i=1}^{\infty} P(A_i)$ can be thought of as an upper bound for $P\left(\bigcup_{i=1}^{\infty} A_i\right)$

Special case:

$P(A \cup B) \leq P(A) + P(B)$

8. *Bonferroni Inequality*:

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq 1 - \sum_{i=1}^{n} [1 - P(A_i)]$$

Special case:

$P(A \cap B) \geq P(A) + P(B) - 1$

$\longrightarrow$ Note that 7) and 8) are useful to bound probabilities.

3. Basic Calculations of Probability

When $|S|$ is finite and has equally likely outcomes, then if $A \subseteq S$ (discrete variable):

$$P(A) = \frac{|A|}{|S|}$$

Example:

Toss 2 coins:

$$S = \{HH, HT, TH, TT\}$$

Let $A =$ getting exactly one head:

$$A = \{HT, TH\}$$

$$P(A) = \frac{2}{4} = \frac{1}{2}$$

Note: Not always easy to count outcomes. See, for example, Section 1.2.3. in Casella and Berger (2002): "Combinatorics" (branch of mathematics that studies counting).